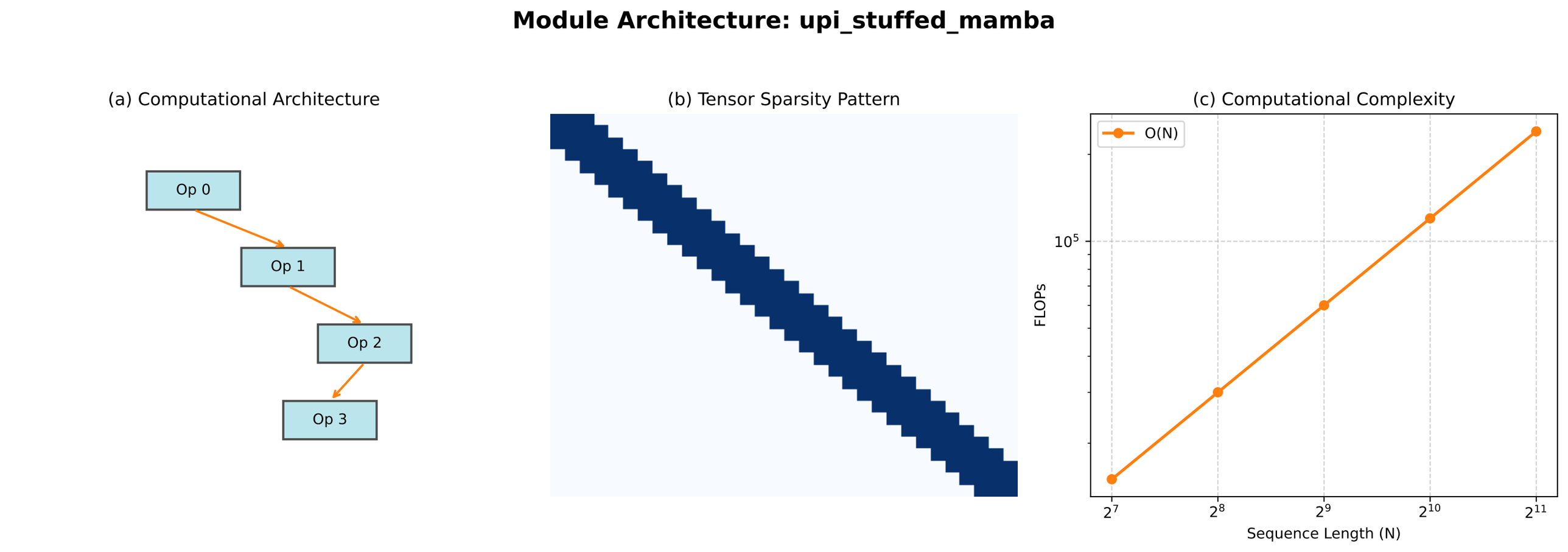Module Architecture: upi_stuffed_mamba